



GARBAGE TO ROAD + Kabadiwala Connect

- Smart Roads for India -



PS ID
229

Category: Software | Theme: Clean & Green Technology (Waste Management)

Team: Shivukumar and Team -  AITM Belagavi | 6 Members



STEP 1

Kabadiwala Cart collection

Doorstep waste collection from
households & shops



STEP 2

QR Scan • No-Text App • Voice

Kannada • Hindi voice support,
simple QR scan



STEP 3

Direct UPI • Instant Payment

No middleman • Secure payment
to Kabadiwala



STEP 4

P1 • P2 • P3 Road Process

P1: Hub → P2: Shredding → P3: Mix
8-10% plastic with Bitumen



Clean India



Strong Roads



Income 2X

"Garbage is not waste — it's a valuable resource" — Dr. Vasudevan

Lat: 15.8497°N, 74.4977°E • Belagavi, Karnataka • Sustainable Infrastructure for India



! PROBLEM garbage plastic waste



Uncollected plastic garbage polluting roads & waterways



Informal waste segregation, low recycling rates



Environmental impact, clogged drains, loss of recyclable value

2 PROBLEM PROOF: Dangerous Roads Across India | Pune to Bengaluru | Public Protesting from earlier



Photo 1 — Waterlogged Bus Stop (Swarajya)



Photo 2 City Pothole Closeup (Deccan Herald)



Photo 3 Dangerous Pothole Accident Spot (The Hindu Express)

3 PROBLEM PROOF: The Exploitation — Middleman vs Kabadiwala



Kabadiwala: Low Earning



Middleman: High Profit



SOLUTION — 4 PILLARS



P1: Collection
QR UPI
Pay for
Waste

Scan waste drop,
instant UPI
reward to user



P2: Aggregation
Segregation &
Storage

Local hub
aggregates, sorts
plastics by type



P3: Shredding &
Processing
Shred to
Aggregate

Plastic shredded into
processed flakes
for mix



P4: Road Construction
with MoRTH
Plastic-Bitumen

Processed plastic
used in roads per
MoRTH guidelines



Innovative No-Text Voice UI



Voice Interface:
No Literacy Barrier

Kannada: "ನಾನು ಕಚರ ತೂರಿಸಿ UPI ಪಡೆದುಲು"

Hindi: "कचरा डालो, UPI भुगतान प्राप्त करो"

- ✓ Voice in Kannada & Hindi for rural users
- ✓ Offline-enabled, works on basic phone via IVR call
- ✓ Audio confirmation for collection & payment



Technologies to be used



Frontend: Flutter

No-Text Voice UI (Kannada & Hindi)



Backend: Firebase & Node.js



Database: Firestore



Payment: UPI API (NPCI)



QR Code System



**Hardware: IoT Weighing Machine +
Plastic Shredder**

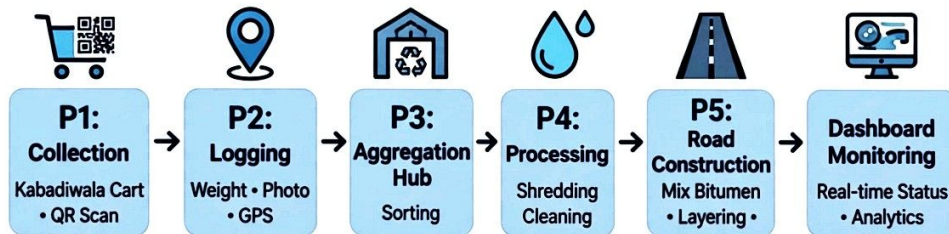


**Road Technology: Plastic-Bitumen Mixing
MoRTH IRC SP 98 | 8-10% ratio**

MoRTH IRC SP 98 | 8-10% ratio



Methodology and process for implementation





Analysis of the feasibility of the idea

GARBAGE TO ROAD + Kabadiwala Connect:

- ✓ Feasibility - Low cost Flutter app + Firebase
- ✓ MoRTH approved plastic road tech
- ✓ Existing kabadiwala network
- ✓ UPI infrastructure ready
- ✓ Belagavi pilot feasible

Feasibility Overview



Potential challenges and risks



Strategies for

overcoming these challenges

- Kabadiwala adoption - low literacy
- Plastic quality variation
- Payment trust
- Logistics
- ✓ No-Text Voice app — Kannada & Hindi training
- ✓ Quality check at P1 hub
- ✓ Instant UPI + digital receipt
- ✓ Partnerships with municipality & PWD



Potential impact on the target audience

- Households
- Kabadiwalas / Informal Workers
- Municipality
- PWD (Public Works Department)
- Citizens

— *GARBAGE TO ROAD + Kabadiwala Connect*



Benefits of the solution (social, economic, environmental, etc.)



Social:

- Doubles kabadiwala income
- Enhances dignity for informal workers
- No-Text empowerment for women & community



Economic:

- Low-cost roads — ~8% cheaper construction
- UPI-based transparent payments
- Creates local jobs for collection & processing



Environmental:

- Reduces landfill waste by ~70%
- Recycles plastic into P1 / P2 / P3 grade roads
- Pothole-free, water-resistant roads per MoRTH standards



These are conditions of Bengaluru roads as mentioned by Deccan Herald, The Hindu Express and Swarajya in their news feed



Photo 1
Waterlogged Bus Stop (Swarajya)



Photo 2
City Pothole Closeup (Deccan Herald)



Photo 3
Dangerous Pothole Accident Spot
(The Hindu Express)

Our solution: garbage-mix road can fix all 3.



Details / Links of the reference and research work:



1. Dr. R. Vasudevan Plastic Road Technology - IIT Madras

Research on using shredded plastic waste in road construction.

<https://iit.ac.in/plastic-road-research>



2. MoRTH Guidelines for use of waste plastic in road construction – IRC SP 98

Ministry of Road Transport & Highways



3. CPCB Report on plastic waste management - Central Pollution Control Board, 2023.

<https://cpcb.nic.in/plastic-waste-management>



4. Flutter Firebase documentation- Official Google documentation for Flutter & Firebase.

<https://firebase.google.com/docs/flutter>

5. UPI API NPCI -Unified Payments data – Municipal solid waste records, 2024-25.

7. News references: Swarajya, Deccan Herald, The Hindu Express – Coverage of Bengaluru pothole crisis.

Sources: Swarajya.com • DeccanHerald.com • TheHinduExpress.com

